

LS-RBF Dataset Selection with Parameter-Mesh Jacobian Checks

Goal

Improve the LS-based primary-mode selection for PROM-RBF with $n_p = 3$ by adding geometric diagnostics:

- Jacobians in a meshed parameter domain,
- Jacobians of the mapped (deformed) reduced domain,
- macro deformation admissibility through $\det(F)$.

1. LS Primary-Mode Construction

Given fluctuation snapshots $W \in \mathbb{R}^{N \times n_f}$ and POD basis $\Phi \in \mathbb{R}^{n_f \times r}$:

$$Q = W\Phi. \quad (1)$$

For $n_p = 3$, targets are chosen as

$$\mu = [E_{xx}, E_{yy}, \gamma_{xy}] \in \mathbb{R}^{N \times 3}. \quad (2)$$

Then LS solves

$$C = \arg \min_C \|QC - \mu\|_F, \quad (3)$$

and builds candidate vectors

$$\Phi_c = \Phi C. \quad (4)$$

The final LS-primary basis is the orthonormalized subspace

$$\Phi_p^{LS} = \text{orth}(\Phi_c), \quad (5)$$

with secondary basis from the orthogonal complement in $\text{span}(\Phi)$.

2. Deformation-Factor Matrix (Parameter Scaling)

To reduce anisotropy in parameter coordinates, define

$$\xi = D_\mu(\mu - \mu_c), \quad D_\mu = \text{diag} \left(\frac{2}{\Delta\mu_1}, \frac{2}{\Delta\mu_2}, \frac{2}{\Delta\mu_3} \right), \quad (6)$$

where $\mu_c = (\mu_{\max} + \mu_{\min})/2$ and $\Delta\mu = \mu_{\max} - \mu_{\min}$.

Interpretation:

- D_μ is a scaling/preconditioning matrix,
- it maps the box-like parameter domain to an approximately isotropic cube,
- it improves numerical robustness for meshing and local Jacobian estimates.

3. Parameter Mesh and Local Jacobians

Build a tetrahedral mesh (Delaunay) in the scaled parameter cloud ξ . For each tetrahedron e with 4 node pairs (μ_i, q_i^{LS}) :

$$M_e = [\mu_1 - \mu_0 \ \mu_2 - \mu_0 \ \mu_3 - \mu_0], \quad Q_e = [q_1 - q_0 \ q_2 - q_0 \ q_3 - q_0]. \quad (7)$$

Local affine map:

$$q^{LS} \approx A_e \mu + b_e, \quad A_e = Q_e M_e^{-1}. \quad (8)$$

Diagnostics:

- $\det(M_e)$: parameter-cell Jacobian (degeneracy check),
- $\det(A_e) = \det(\partial q^{LS} / \partial \mu)$: local orientation/stretch from parameter domain to LS domain.

Negative $\det(A_e)$ indicates local orientation inversion of the mapping.

4. Deformed-Domain Jacobian Check in Mechanics

For each applied Green-Lagrange strain E , compute

$$C = 2E + I, \quad F = C^{1/2}, \quad J_{phys} = \det(F). \quad (9)$$

Admissibility requires $J_{phys} > 0$. If $J_{phys} \leq 0$, deformation is non-physical (inversion).

5. Practical Benefit for RBF with $n_p = 3$

These checks help identify whether the reduced coordinate map is locally well-oriented and non-degenerate, which is often a hidden cause of instability or poor regression quality in compact-center RBFs.

6. Files Produced by the Implementation

The updated stage saves:

- parameter mesh nodes and tetrahedra,
- Jacobian arrays ($\det(J_\mu)$, $\det(\partial q^{LS} / \partial \mu)$),
- macro $\det(F)$ values,
- histogram plots,
- text and `.tex` summaries.